



PTT PHENOL COMPANY LIMITED

PROCESS SPECIFICATION

**CALORIMETER NO.1/NO.2
(X-2309A/B)**

Docu. No. : 14780-8120-PS-X2309

Project No. : 120117

Rev. No. : Z1

Date : MAY 10, 2016

Page : 1 / 4

TITLE :

CALORIMETER NO.1/NO.2

(X-2309A/B)

DISCIPLINE : PROCESS

DOCUMENT NO. : 14780-8120-PS-X2309

PROJECT NAME : PTT Phenol Train II Project

PROJECT NO. : 120117

OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 29, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	DEC 11, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 2, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUL 19, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREP'N	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

This Document is the property of POSCO Engineering Co., Ltd. Therefore, it shall not be released to Any third party without permission of an authorized personnel of the POSCO Engineering Co., Ltd.

[illegible]

POSCO ENGINEERING

PROCESS SPECIFICATION

CLIENT	: PTT PHENOL	REV.	DATE	REV.	DATE
JOB NAME	: PTT PHENOL TRAIN II PROJEC T	A1	OCT 2, 2013		
JOB NO	: 120117	A2	DEC 11, 2013		
PLANT LOCATION	: MAP TA PHUT, THAILAND	F1	MAY 29, 2014		
REV DOC. NO	: 14780-8120-PS-X2309	Z1	MAY 10, 2016		

Item Number : X-2309A/B (P&ID No.: 14780-8120-25-23-0018)

Service : Calorimeter No. 1 and No. 2

Required Quantity : 2 (Two)

Operating Condition

Fluid : Hydrocarbon and Sulfuric Acid

Operating Temperature : 79 °C

Operating Pressure : 2.1 kg/cm²g

Mechanical Design Data

Shell diameter, inch : 4

Pipe Schedule/Thickness : Schedule 40

Design Pressure, kg/cm²(g) : 10.5

Design Temperature, °C : 250

Material of Construction

Flanges : A182 GR. F 304L

Forgings : A182 GR. F 304L

Fittings : A403 GR. WP 304L

Pipe : A312 GR. TP 304L Seamless

Gaskets : Spiral Wound per API 601

Bolting : Stud bolts A193-B7 Threaded full length with two A194 GR. 2H Nuts

Plate : A240 type 304L SS

Internal : Hastelloy C276

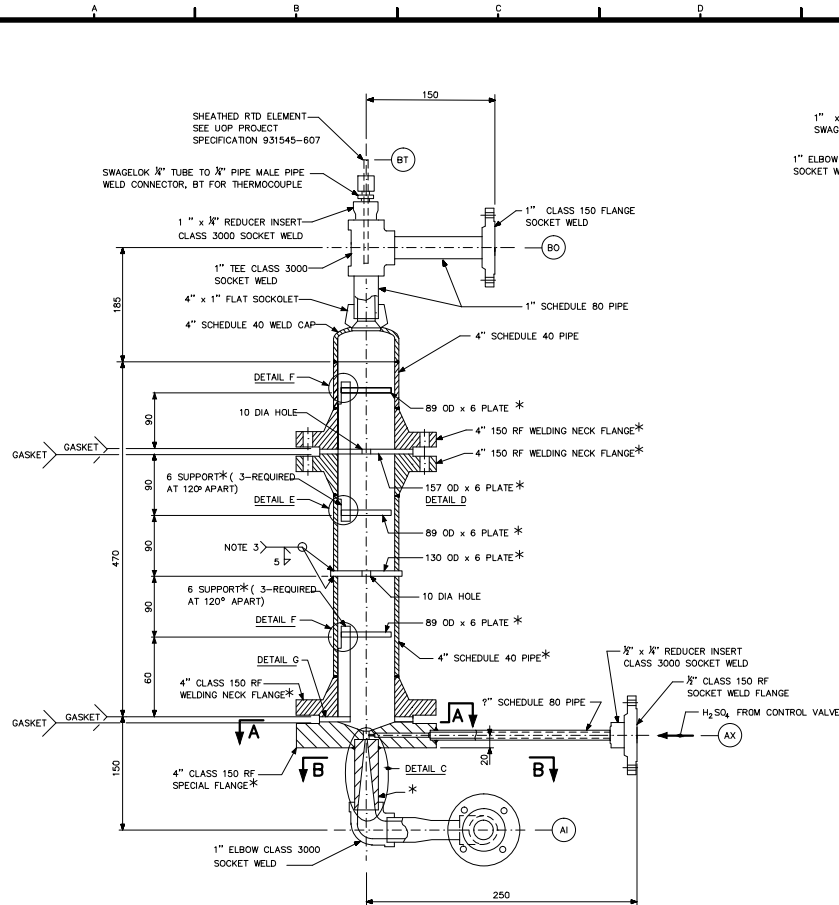
Nozzle Connection.

Mark	Qty.	Size	Class	Service	Remark
AI	1	1/2"	150	Feed Inlet	See Calorimeter Assembly and details
AX	1	1/2"	150	H2SO4 Inlet	See Calorimeter Assembly and details
AT	1	1/4"		RTD Connection	See Calorimeter Assembly and details
BO	1	1"	150	Outlet	See Calorimeter Assembly and details
BT	1	1/4"		RTD Connection	See Calorimeter Assembly and details

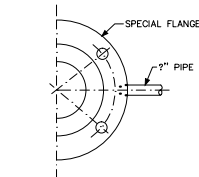
Attachment

Calorimeter Assembly and Details

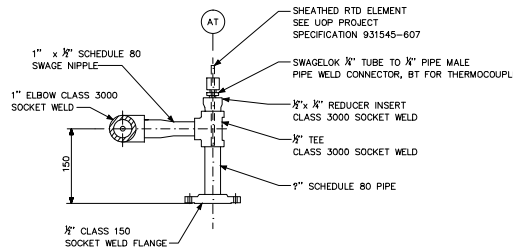
THIS P AND I DIAGRAM IS DIAGNOSTIC ONLY. DESIGN OF PIPE LINES MUST BE INVESTIGATED FOR VENTING OF GAS AND VAPOR POCKETS IN PIPING AND EQUIPMENT. LOW POINTS IN PIPING, PUMPS AND EQUIPMENT FOR FREEZING AND DRAINING, AND ACCESSIBILITY OF ALL VALVES, FLANGES, INSTRUMENTS INCLUDING THERMOCOUPLES, ETC.



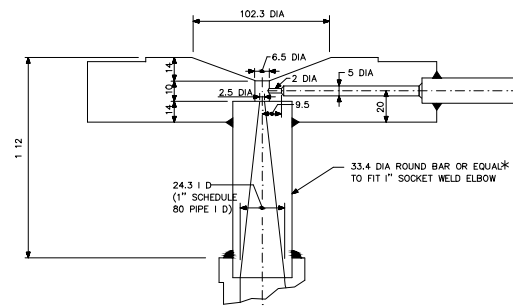
ELEVATION



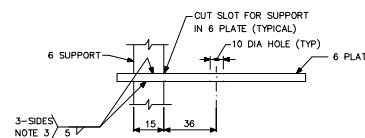
VIEW A-A



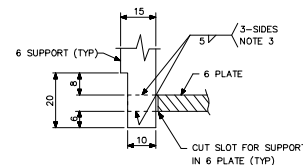
SECTION B-B



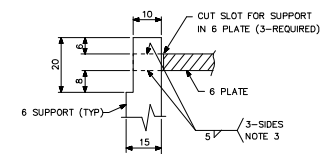
DETAIL C



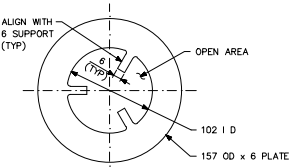
DETAIL D



DETAIL E

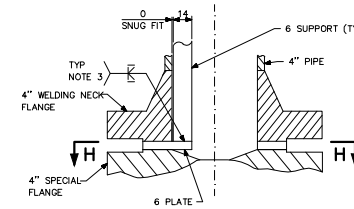


DETAIL F



SECTION H-H

PLATE ONLY



DETAIL G

NOTES:

1. FABRICATE IN ACCORDANCE WITH CHEMICAL PLANT AND PETROLEUM REFINERY PIPING CODE ASME B31.3.
2. INSTALL IN VERTICAL POSITION.
3. ALL WELDS INSIDE OF CALORIMETER TO BE GROUND SMOOTH.
4. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS NOTED.

MATERIAL PER ASTM SPECIFICATIONS

- FLANGES : A182 GR. F304L
 FORGINGS : A182 GR. F304L
 FITTINGS : A403 GR. WP304L
 PIPE : A312 GR. TP304L SEAMLESS
 GASKETS : SPIRAL WOUND PER API 601
 BOLTING : STUD BOLTS A193-B7 THREADED FULL LENGTH WITH TWO A194 GR. 2H NUTS.
 PLATE : A240 TYPE 304LSS
 * MATERIAL : HASTELLOY C276

REV	DATE	BY	CHK	DATE	REV	DATE	BY	CHK	DATE
1					1				
UOP A Honeywell Company PHENOL Calorimeter Assembly and Details ITEM X-2309A/B CONCENTRATION, DECOMPOSITION AND NEUTRALIZATION SECTION UOP PHENOL PROCESS UNIT PTT PHENOL COMPANY LIMITED (PPCL) UNIT 2 HEMARAJ EASTERN INDUSTRIAL ESTATE MAP TA PHUT, RAYONG, THAILAND									
DATE	CHK	APVD	DATE	PROJECT	TYPE	DWG NO.	REV		
MMW	MYW	MYW	12	16APR		963766-301-99-A1	0		